Fiscal Year:	FY 2005	**Task Last Updated:**	FY 09/27/2006
PI Name:	Faeth, Gerard Michael Ph.D.		
Project Title:	Investigation of Laminar Jet Diffusion Flames in Microgravity: A Paradigm for Soot Processes in Turbulent Flames		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	gmfaeth@umich.edu	**Fax:**	FY 734-936-0106
PI Organization Type:	UNIVERSITY	**Phone:**	734 764-7202
Organization Name:	University of Michigan		
PI Address 1:	3000 FXB Bldg., 1320 Beal Avenue		
PI Address 2:			
PI Web Page:	www.engin.umich.edu/dept/aero/people/faculty/gmfaeth.html		
City:	Ann Arbor	**State:**	MI
Zip Code:	48109-2140	**Congressional District:**	15
Comments:	Faeth, Gerard M.--DECEASED Jan 2005 (per N. Hall/GRC)		
Project Type:	Flight	**Solicitation / Funding Source:**	OTHER
Start Date:	05/01/1997	**End Date:**	03/31/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	Urban, David	**Contact Phone:**	216-433-2835
Contact Email:	david.l.urban@nasa.gov		
Flight Program:			
Flight Assignment:	note - date changed from 12/02/2002		
Key Personnel Changes/Previous PI:	Faeth, Gerard M.--DECEASED Jan 2005 (per N. Hall/GRC)		
COI Name (Institution):			
Grant/Contract No.:	NAG 3 2404		
Performance Goal No.:	1.22		
Performance Goal Text:	Enable increased combustion system efficiency, reduced pollution, and mitigation of fire risks through insights and data bases obtainable only through microgravity experiments.		

Task Description:	The structure and the soot formation and oxidation properties of laminar nonpremixed and premixed flames are being studied both experimentally and computationally; conditions where the intrusion of buoyancy is small are emphasized so that practical nonbuoyant flames can be approximated by exploiting classical laminar flamelet concepts. The investigation is divided into three phases, considering ground-based laminar premixed and nonpremixed flames and space-based laminar nonpremixed flames, respectively. This flight project is an experimental and theoretical investigation of the mechanisms of soot formation in laminar jet diffusion flames under conditions of low buoyancy. The flight experiment is in the Combustion Experiment Modules 1 and 2 (CM-1 and CM-2). In most flames of practical interest, soot radiation is the dominant mode of heat transport to combustor components and a dominant mechanism for flame spread and growth. The experimental results, combined with theoretical modeling, will either confirm or deny the applicability of the conserved scalar formalism to soot properties in diffusion flames. These results, when combined with those of ground-based low- and normal- gravity experiments will increase the current fundamental understanding of soot formation processes in both laminar and turbulent flames and increase our ability to model them. Consequently this will have a significant impact on our understanding of systems such as the spread of unwanted fires, design of jet engines and large scale boilers.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress report received this period, due to death of PI.
Bibliography Type:	Description: (Last Updated: 10/19/2009)